WEEK 1 · LESSON 2 OF 24 · HANDOUT

How Machines Learn: Data, Patterns and Models

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=R9OHn5ZF4Uo — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 4: Learning (supervised, reinforcement, unsupervised) — https://www.youtube.com/watch?v=E4M_IQG0d9g

Your goal in this lesson: Describe the three steps of machine learning (data, training, prediction), explain why data quality matters, and name the three main types of learning.

Learning by example — just like people

How did you learn to recognise a mango? Nobody gave you a rule like "oval, 10–15 cm, green-yellow". You saw many mangoes, and your brain found the pattern. Machine learning works the same way, only with far more examples and mathematics instead of a brain.

The three steps of machine learning

1. **Data** — collect many examples. For a model that predicts crop yields, that could be past rainfall, soil type, fertiliser used and the harvest that followed.
2. **Training** — the computer looks at the examples again and again, adjusting itself to make fewer mistakes each time. The result is a **model**.
3. **Prediction** — the model is given something new it has never seen (this season's rainfall and soil) and makes a prediction (the expected harvest).

A simple way to remember it: **examples in → pattern learned → prediction out.**

Features and labels

- A **feature** is a piece of information the model looks at: rainfall, house size, number of rooms.
- A **label** is the answer we want the model to learn to predict: the harvest, the house price, "spam" or "not spam".

Feature: house size	Feature: rooms	Feature: area	Label: monthly rent
80 m²	3	Kicukiro	250,000 RWF
120 m²	4	Kacyiru	450,000 RWF
60 m²	2	Nyamirambo	150,000 RWF

After seeing thousands of rows like these, a model can estimate the rent of a house it has never seen.

Three ways machines learn

- **Supervised learning** — learning from examples *with* the correct answers (labels). Example: thousands of X-ray images each marked "pneumonia" or "healthy".

- **Unsupervised learning** — finding groups in data *without* answers. Example: a supermarket discovering that its customers fall into four shopping styles.
- **Reinforcement learning** — learning by trial, error and reward. Example: a program that learns to play a game by getting points for winning.

Garbage in, garbage out

A model can only be as good as its data.

- If the data is **too small**, the model has not seen enough to learn the real pattern.
- If the data is **wrong or messy**, the model learns mistakes.
- If the data is **unbalanced** — for example, a face-recognition system trained mostly on light-skinned faces — the model will work worse for the people it rarely saw. This is one way **bias** enters AI, and we return to it in Week 4.

Why this matters for you

When you use an AI tool, remember it is drawing on patterns from its training data. That explains both its strengths (it has "seen" a lot) and its weaknesses (it may not know recent events, local facts about Rwanda, or your organisation's own information unless you give them to it).

Remember this: Machine learning turns examples into a model that makes predictions — and the quality of the data decides the quality of the model.

Name: _____ Date: _____

Activity

Task 1

Write the **label** you want to predict.

Task 2

List **five features** that could help predict it.

Task 3

Write one way the data could be **biased or misleading**, and how you would fix it.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. What are the three basic steps of machine learning, in order?
   A. Coding, testing, selling
   B. Training, prediction, data
   C. Prediction, data, training
   D. Data, training, prediction
2. In a model that predicts house rent, which of these is the LABEL?
   A. Number of rooms
   B. Neighbourhood
   C. Monthly rent
   D. House size

3. Learning from examples that already have the correct answers is called…

A. Unsupervised learning
B. Supervised learning
C. Reinforcement learning
D. Random learning

4. A supermarket groups its customers into shopping styles without any labels. This is…

A. Unsupervised learning
B. Reinforcement learning
C. Not machine learning
D. Supervised learning

5. What does 'garbage in, garbage out' mean for AI?

A. A model can only be as good as the data it learned from
B. AI throws away old data
C. Computers need regular cleaning
D. AI cleans dirty data automatically

6. Why might an AI tool not know a recent local event in Rwanda?

A. Because it only knows patterns from its training data, which may not include it
B. Because the event was too small to matter
C. Because AI cannot read English
D. Because AI is not allowed in Rwanda

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